

TCP (Transmission Control Protocol) is a **connection-oriented and reliable transport layer protocol**. It is used when data must reach the destination **accurately, completely, and in the correct order**. Before sending any data, TCP first establishes a connection between the client and the server using the **Three-Way Handshake (SYN, SYN-ACK, ACK)**. During communication, TCP uses **sequence numbers** and **acknowledgement numbers** to track data and confirm successful delivery. If any packet is lost, TCP retransmits it, making communication reliable. In this practical, we will use **Wireshark** to capture TCP packets while uploading a file from our computer to a remote server. Then we will analyze how TCP establishes the connection and transfers the data step by step.

Three-Way Handshake Process

1. **SYN (Synchronize):**
The **client** sends a **SYN** packet to the server requesting to establish a connection.
2. **SYN-ACK (Synchronize-Acknowledge):**
The **server** replies with a **SYN-ACK** packet, indicating that it has received the client's request and is ready to establish the connection.
3. **ACK (Acknowledge):**
The **client** sends an **ACK** packet to confirm the server's response. After this, the TCP connection is successfully established and data transfer begins.

Use of Sequence Number

- A **Sequence Number** is assigned to each TCP packet.
- It helps maintain the **correct order** of packets at the receiver.
- If a packet is lost, TCP can identify the missing packet and retransmit it.

Use of Acknowledgement Number

- An **Acknowledgement Number** is sent to give confirmation that packets have been received successfully.
- It indicates the **next sequence number expected** from the sender.
- If an acknowledgement is not received, TCP retransmits the missing packet, ensuring reliable communication.

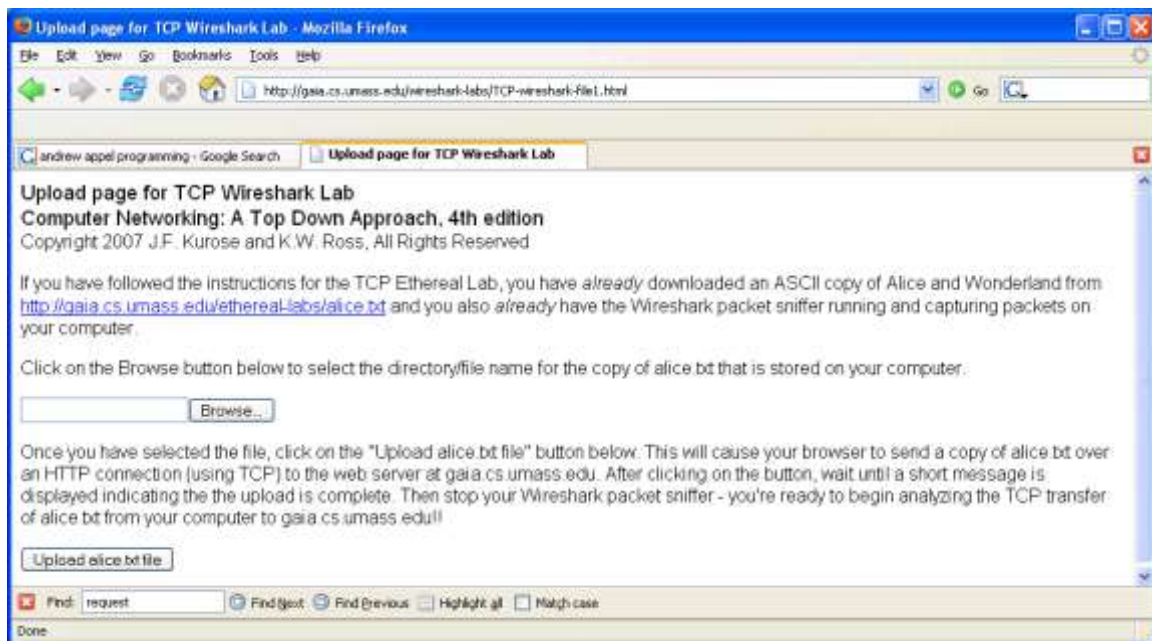
In this practical, we will study the working of the TCP (Transmission Control Protocol) using Wireshark. First, we will capture TCP packets while uploading a file from our computer to a remote server using the HTTP POST method. After capturing the packets, we will analyze different TCP features such as the TCP connection, Three-Way Handshake (SYN, SYN-ACK, ACK), sequence numbers, acknowledgement numbers, and data transfer. This practical helps us understand how TCP establishes a reliable connection and transfers data between the client and the server.

1. Capturing a bulk TCP transfer from your computer to a remote server

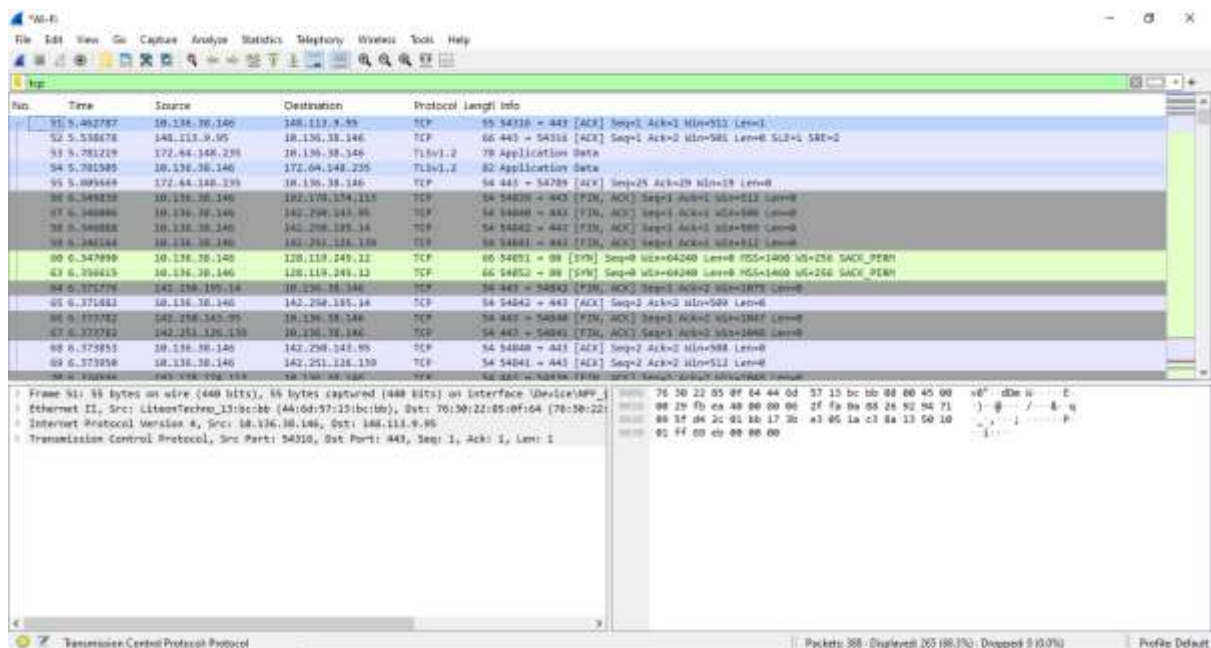
Before we start learning TCP, we need to capture TCP packets using Wireshark. For this, we will first download a file called `alice.txt`. This file contains the ASCII text of the book *Alice in Wonderland*. Then, we will open the given web page, select the `alice.txt` file from our computer, and upload it to a remote web server using the HTTP POST method. We use the POST method because it is used to send a large amount of data from the client to the server. While the file is being uploaded, Wireshark will capture all the TCP packets sent and received between our computer and the server. Later, we will analyze these captured TCP packets.

Do the following:

- Start up your web browser. Go the <http://gaia.cs.umass.edu/wireshark-labs/alice.txt> and retrieve an ASCII copy of *Alice in Wonderland*. Store this file somewhere on your computer.
- Next go to <http://gaia.cs.umass.edu/wireshark-labs/TCP-wireshark-file1.html>



- Use the *Browse* button in this form to enter the name of the file (full path name) on your computer containing *Alice in Wonderland* (or do so manually). Don't yet press the "*Upload alice.txt file*" button.
- Now start up Wireshark and begin packet capture (*Capture->Start*) and then press *OK* on the Wireshark Packet Capture Options screen (we'll not need to select any options here).
- Returning to your browser, press the "*Upload alice.txt file*" button to upload the file to the gaia.cs.umass.edu server. Once the file has been uploaded, a short congratulations message will be displayed in your browser window.
- Stop Wireshark packet capture. Your Wireshark window should look similar to the window shown below.



Before analyzing the TCP connection in detail, let's first get an overview of the captured packets. First, apply the `tcp` filter in the Wireshark Display Filter bar (type `tcp` in lowercase and press Enter). This will display only TCP packets, making them easier to analyze.

Now check whether your capture contains these packets. First, look for the SYN packet, which indicates the start of the TCP connection (Three-Way Handshake). Then, look for the HTTP POST packet, which is used to upload the `alice.txt` file. If you can see these packets, it means the packet capture was successful.

Answer the question.

1. What is the IP address and TCP port number used by the client computer (source) that is transferring the file to gaia.cs.umass.edu?

To answer this question, the easiest way is to select the HTTP POST message (or any HTTP message of the same file upload) in Wireshark. Then, open the TCP packet details. In the packet details window, you can find the client's IP address and TCP port number.

2. What is the IP address of gaia.cs.umass.edu? On what port number is it sending and receiving TCP segments for this connection?

3. What is the IP address and TCP port number used by your client computer (source) to transfer the file to gaia.cs.umass.edu?

4. What is the sequence number of the TCP SYN segment that is used to initiate the TCP connection between the client computer and gaia.cs.umass.edu? What is it in the segment that identifies the segment as a SYN segment?

(match this things for selecting a syn packet, this are present in http post packet)

Ip of server: 128.119.245.12

source port :54851 (this is client port num)

Destination port :80 (this is server port num)

->sequence num:0

->Flags: 0x002 (SYN)

5. What is the sequence number of the SYNACK segment sent by gaia.cs.umass.edu to the client computer in reply to the SYN? What is the value of the Acknowledgement field in the SYNACK segment? How did gaia.cs.umass.edu determine that value? What is it in the segment that identifies the segment as a SYNACK segment?

Sequence Number: 0 (relative). The Sequence Number of the SYN-ACK packet is 0.

Acknowledgement Number: 1 (relative). The Acknowledgement Number of the SYN-ACK packet is 1.

ACK = 1 because the client sent a SYN packet with Sequence Number 0. The server acknowledges the next expected Sequence Number, so it sends ACK = 1.

The packet is identified as a SYN-ACK packet because the TCP Flags field contains both SYN and ACK flags (Flags = SYN, ACK / 0x012).

6. Find the TCP packet that contains the HTTP POST command. What is its Sequence Number?

To find the HTTP POST command, find the packet whose Data field contains the word "POST".

7. What is the length of each of the first six TCP segments?